

Introduction

Hirshfeld atom refinement (HAR) [1, 2] is an advanced approach to X-ray crystal structure refinement. It is a post-IAM (IAM = Independent Atom Model) refinement procedure. HAR is originally developed by D. Jayatilaka and available with the program Tonto. After its first lunch on 2008 different software implementations have been developed, starting by using for example the standalond HART (terminal version of Tonto to uniquely run HAR) which was interfaced inside Olex2. Currently, Olex2 also offers HAR using its own olex2.refine software in combination with NoSpherA2 and a quantum chemistry calculation software of your choice (such as Gaussian, or Orca), the reason why so many softwares may be required to run such a refinement procedure should be clear after the following explanation.

Within softwares such as SHELXL, the refinement procedure is based on the minimization of:

$$M = \sum w(|F_o| - |F_c|)^2$$

Where, F_o and F_c corresponds to observed and calculated structure factors respectively. These, however, rely on the sum of the form factors f_j :

$$F(\vec{h}) = \sum_{j=1}^N f_j \cdot e^{2\pi i \vec{h} \cdot \vec{r}}$$

These form factors f_j are basically a mathematical expression of how an **isolated atom type** will scatter incoming radiation, and for X-rays in SHELXL these are tabulated and listed under the famous Tables 4.2.6.8 and 6.1.1.4 of the International Tables Vol C.

Now be aware of the bold statement written on the sentence above, and its implication: every atom is modeled as an isolated non-interacting sphere, located at the center of its electronic cloud, independent of its bonding environment or oxidation state. Also notice, that this would place hydrogen atoms at its maxima of electron density, which is located in the bond instead of around its nuclei, leading to the infamous shortening of X-H bond lengths in X-ray diffraction studies.

In HAR, these form factors are not read from the ITC, instead, they are calculated on the fly for each atom (! Not atom type!) based on a Hirshfeld partitioning of the molecular electron density. This molecular electron densities (ED) on the other hand, are also obtained on the fly by performing the best reasonably achievable *ab-initio* quantum mechanical calculations (which relies on a choice of method and basis set) for your molecule.

The use of the Hirshfeld partitioning will lead to atomic electron densities (and consequently form factors), which are aspherical and deformed according to the environment of each atom, accounting for aspherical effects, e.g. caused by bonding.

The sequence of how the refinement and these steps are performed are explained below.

Metodology

Please notice that we rely on an initial geometry, therefore, HAR can be considered a post-IAM procedure.

Based on the initial IAM geometry, the steps involved in HAR are:

1 - A theoretical single-point calculation which provides the molecular electron-density distribution.

2 - This molecular electron density undergoes:

- (i) a stockholder partitioning according to Hirshfeld [3, 4] to obtain the aspherical atomic electron densities;

$$\rho(A) = w_A(\vec{r}) \cdot \rho_{molecule}(\vec{r}), \quad w_A(r) = \frac{\rho_A^0(\vec{r} - \vec{r}_A)}{\sum_B \rho_B^0(\vec{r} - \vec{r}_B)}$$

Aspherical density of an atom at its position $\vec{r}_A$ in a molecule. Hirshfeld or stockholder weight function where $\rho_B^0(\vec{r})$ is the spherically averaged atomic density for the isolated atom B centered at the origin.

- (ii) aspherical atomic electron densities are thermally smeared (by the convolution with the probability density function) and;

$$\langle \rho(A) \rangle = \rho_A * P_A$$

Thermally smeared aspherical atomic electron density.

- (iii) Fourier transformed to provide tailor-made aspherical atomic scattering factors for the molecule being studied.

$$F_j = \sum_a^N \sum_{\vec{r}} n_A^{-1} \int \langle \rho_a \rangle(\vec{r}') e^{i\vec{q}_j \cdot \vec{r}'} d\vec{r}'$$

Aspherical structure factor, where N are the unique atoms that constitute an asymmetric unit. The factor n_A is a site-symmetry factor, i.e. the number of times the unique atom A is mapped onto itself.

3 - A full-matrix least-squares refinement of the parameters is then performed using the newly calculated aspherical scattering factors in order to reproduce the observed X-ray diffraction data.

At the end of each refinement cycle, the geometry of the molecule changes and therefore, the initial molecular electron density should be updated. Therefore, these steps are repeated until full convergence is achieved (in energy and geometry).

Since the wavefunction calculated in the energy computation in Tonto is based on an isolated molecule, the influence of the environment (or crystal effects) would not be reflected in the molecular electron density. Therefore, self-consistent atomic charges

and dipoles of the Hirshfeld atoms are calculated and used to surround the isolated molecule for which the wavefunction is calculated to simulate a crystalline environment.

The simulation of such effects by point charges and dipoles might be insufficient, e.g. for situations with strong hydrogen bonding, for such cases a more exhaustive treatment which explicitly includes the molecules involved in the strong interactions should be used.

Charges are placed at every atom position which have are located within a specified distance of the central molecule (defined in the software as cluster radius). Complete molecules should be used, to avoid charged species and therefore minimize errors in slowly converging potential sums, but the option to use non-complete molecules is now available¹ and it is useful when dealing with network compounds.

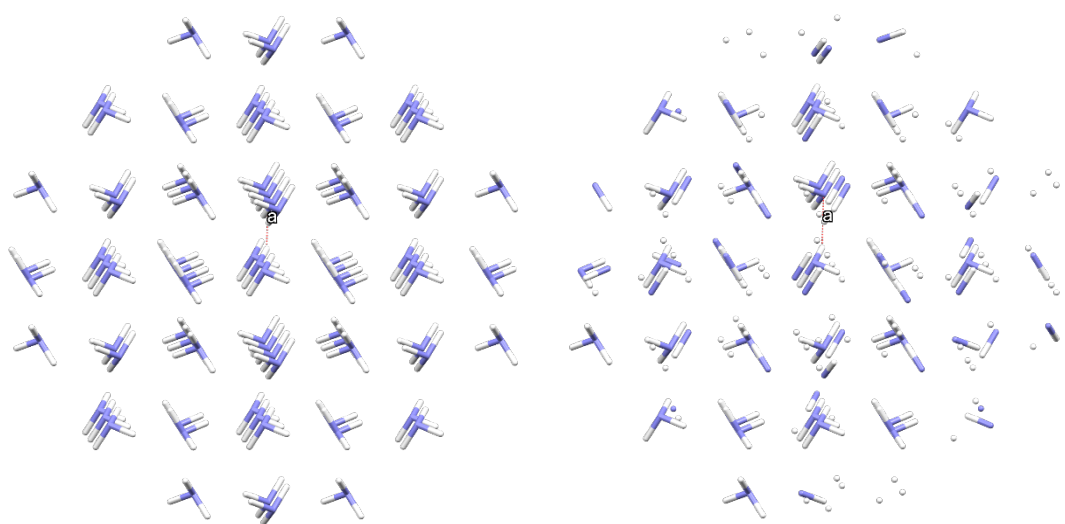


Figure 1: visual demonstration of a cluster of completed molecules on NH₃ around the center molecule vs a cluster not completed. Notice that the molecules in the edge of the sphere radius are “broken”.

It has been shown that including a surrounding cluster of charges is more important than using a large basis set,[2] and results for hydrogen atoms that are comparable to those obtained from neutron diffraction experiments might be achieved.[5, 6] When using HF/def2-SVP not including a self-consistent field of cluster charges surrounding the main molecule, this setting is called the “minimal HAR” [5] and is a good starting point.

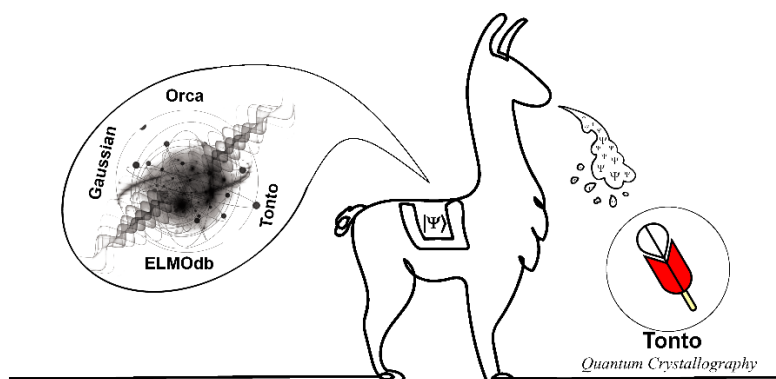
¹ This option was added by us in the Tonto code in Dec 4, 2017 (commit fe66cc2), and therefore does not work with earlier versions of Tonto.

Important facts:

1 - In the current version of Tonto, all hkl files must be merged and pruned of systematic absences.

2 - Since HAR is subjected to a theoretical single-point calculation of a molecular fragment for extraction of aspherical scattering factors, structures with $Z' < 1$ the input geometry in the CIF must be modified in order to include one “full molecule” and/or “strong interactions” that might affect the ED.

3 - During the SCF computation in Tonto, all linear dependencies in the least-squares matrix are eliminated automatically, therefore no restraints or constraints are implemented in the program. This might cause problems in the case of spherical ions. To overcome this problem, we have created a script (lamaGOET – see below) that is able to perform a HAR (iteratively as the original HAR within Tonto) using different softwares for the SCF calculation (step 1), such as Gaussian, Orca², the original Tonto or the Elmo data base (by Alessandro Genoni). The lamaGOET script also provides a graphical interface which facilitates the generation of Tonto input files including advanced options such as X-ray wavefunction fitting and plotting.



2 The implementation with Orca is still not fully finalized since the current version of Tonto still cannot handle spherical basis-sets.

Required softwares

The following softwares are required for the workshop:

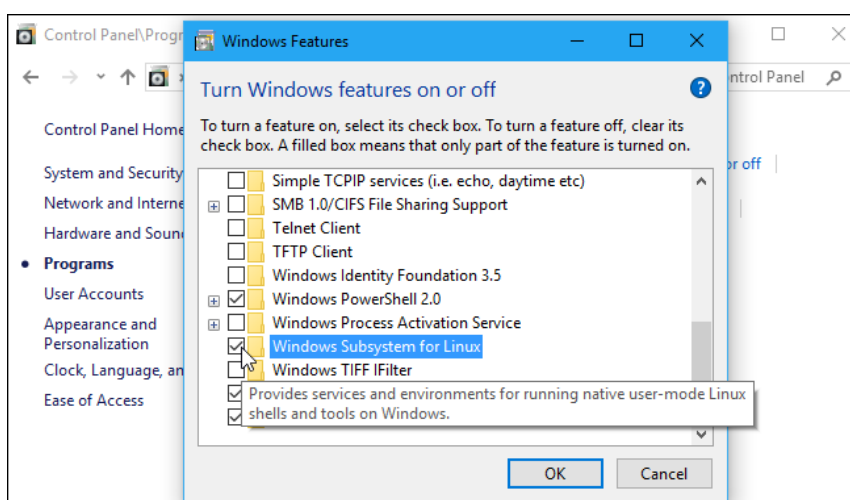
- 1- Tonto
- 2- lamaGOET
- 3- Orca (optional)
- 4- Gaussian (optional)
- 5- ELMOdb (optional)

Installation procedures

Windows

The easiest way to run all the softwares on a windows machine is to use the windows subsystem for linux available for Windows10 and above. (This feature doesn't work on the 32-bit version of Windows 10, so ensure you're using the 64-bit version of Windows.)

Assuming you have 64-bit Windows, to get started, head to Control Panel > Programs > Turn Windows Features On Or Off. Enable the "Windows Subsystem for Linux" option in the list, and then click the "OK" button.

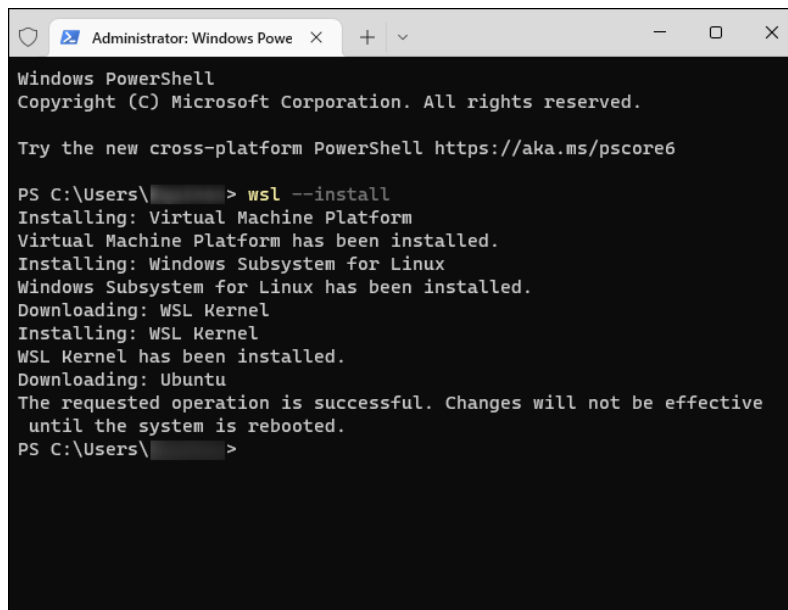


Click "Restart now" when you're prompted to restart your computer. The feature won't work until you reboot.

Alternatively, you can also install it using PowerShell. Launch PowerShell as an Administrator, then enter:

```
wsl --install
```

It'll take a few minutes to download and install all the required components — after it does, you need to restart your computer.



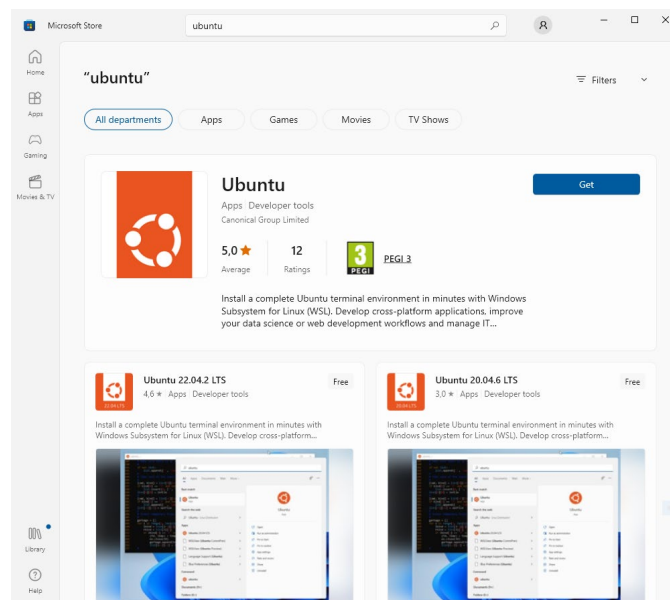
```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\[redacted] > wsl --install
Installing: Virtual Machine Platform
Virtual Machine Platform has been installed.
Installing: Windows Subsystem for Linux
Windows Subsystem for Linux has been installed.
Downloading: WSL Kernel
Installing: WSL Kernel
WSL Kernel has been installed.
Downloading: Ubuntu
The requested operation is successful. Changes will not be effective
until the system is rebooted.
PS C:\Users\[redacted] >
```

After your computer restarts, open the Microsoft Store from the Start menu, and search for “Ubuntu” in the store. Please make sure it is at least version Ubuntu 22.04.

To install a Linux distribution, click it, and then click the “Get” or “Install” button to install it like any other Store application.



You now have a full command-line bash shell based on Ubuntu, or whatever other Linux distribution you installed.

Because they're the same binaries, you can use Ubuntu's *apt* or *apt-get* command to install software from Ubuntu's repositories if you're using Ubuntu. Just use whatever command you'd normally use on that Linux distribution. You'll have access to all the Linux command line software out there, although some applications may not yet work perfectly.

To open the Linux environment you installed, just open the Start menu and search for whatever distribution you installed. For example, if you installed Ubuntu, launch the Ubuntu shortcut.

You can pin this application shortcut to your Start menu, taskbar, or desktop for easier access.

The first time you launch the Linux environment, you'll be prompted to enter a UNIX username and password. These don't have to match your Windows username and password, but will be used within the Linux environment.

For example, if you enter "*bob*" and "*letmein*" as your credentials, your username in the Linux environment will be "*bob*" and the password you use inside the Linux environment will be "*letmein*" — no matter what your Windows username and password are.

On Ubuntu, you need to prefix a command with *sudo* to run it with root permissions. The "root" user on UNIX platforms has full system access, like the "Administrator" user on Windows. Your Windows file system is located at */mnt/c* in the Bash shell environment.

Use the same Linux terminal commands you'd use to get around. If you're used to the standard Windows Command Prompt with its DOS commands, here are a few basic commands common to both Bash and Windows:

Change Directory: *cd* in Bash, *cd* or *chdir* in DOS

List Contents of Directory: *ls* in Bash, *dir* in DOS

Move or Rename a File: *mv* in Bash, *move* and *rename* in DOS

Copy a File: *cp* in Bash, *copy* in DOS

Delete a File: *rm* in Bash, *del* or *erase* in DOS

Create a Directory: *mkdir* in Bash, *mkdir* in DOS

Use a Text Editor: *vi* or *nano* in Bash, *edit* in DOS

It's important to remember that, unlike Windows, the Bash shell and its Linux-imitating environment are case-sensitive. In other words, "File.txt" with a capital letter is different from "file.txt" without a capital.

You'll need to use the *apt* command to install and update the Ubuntu environment's software. Be sure to prefix these commands with *sudo*, which makes them run as root—the Linux equivalent of Administrator. Here are the *apt-get* commands you'll need to know:

Install an Application Package: *sudo apt install packagename* (Replace "packagename" with the package's name.)

Uninstall an Application Package: *sudo apt remove packagename* (Replace "packagename" with the package's name.)

Search for Available Packages: *sudo apt search word* (Replace "word" with a word you want to search package names and descriptions for.)

Download and Install the Latest Versions of Your Installed Packages: `sudo apt upgrade`

After you've downloaded and installed an application, you can type its name at the prompt, and then press Enter to run it. Check that particular application's documentation for more details.

Since we will need to use a graphical interface to run lamaGOET, we need to export the Xserver inside your linux subsystem. There are several ways to do this, the simplest one is using Xming, but Xming usually leads to a lower resolution making the lamaGOET window with a very small text. The other option is by using mobaXterm.

1 - If you chose to use Xming, download and install Xming from the following website:

<https://sourceforge.net/projects/xming/>

After installing Xming, restart your computer and run Xming by searching for the software on the Start menu for windows.

Once the software starts you should see the little X icon next to the windows clock of your computer.

2 - If you choose to use mobaXterm, download and install it from the following website (The free version is enough):

<https://mobaxterm.mobatek.net/download.html>

Once the software starts you should see on the top right corner window the button X server goes from a gray/black to a colorful icon like shown below:



That means that your Xserver is running and available for other applications to use, therefore, now, search for Ubuntu on the Start menu and run it.

Now inside the Ubuntu bash terminal that opened, you can navigate to the folder where you have placed the example files for the workshop by typing:

```
cd /mnt/<c or d>/<Full path to folder>
```

For convenience, we will install Tonto and lamaGOET in the examples folder.

Go to the Linux instructions and continue from there!

Linux

Install the dependencies by running the following commands:

```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get install gfortran
sudo apt-get install g++
sudo apt-get install git
sudo apt-get install liblapack-dev
sudo apt-get install cmake
sudo apt-get install coreutils
sudo apt-get install gawk
sudo apt-get install tofrodos
sudo apt-get install libblas-dev libblas3
```

Now let's install Tonto (<https://github.com/dylan-jayatilaka/tonto>):

```
git clone --recurse-submodules https://github.com/dylan-jayatilaka/tonto.git
```

Once the cloning is finished type the following commands:

```
cd tonto
git checkout release-no-ptr
mkdir build && cd build
cmake .. -DCMAKE_Fortran_COMPILER=gfortran -DCOMPILER_LAPACK=ON
make -j
```

and wait for the compilation to finish. **The compilation should take about 30 minutes, so we advise you do this before the workshop!**

if you get an error message like this:

```
CMake Error at CMakeLists.txt:9 (cmake_policy):
```

```
Policy "CMP0110" is not known to this version of CMake.
```

Please change the file *CMakeLists.txt* located inside the Tonto folder by placing an “#” marker in front of line number 9:

```
#cmake_policy(SET CMP0110 NEW)
```

Try to run the cmake and the make commands shown previously again, and if now you get an error message saying:

```
gfortran-8: error: unrecognized command line option '-fallow-invalid-boz'; did you mean '-Wno-invalid-pch'?
```

Then please edited the file *SetFortranFlags.cmake* located inside the Tonto/cmake folder by deleting the text **-fallow-invalid-boz** in line number 30, from:

```
set(CMAKE_Fortran_FLAGS "${CMAKE_Fortran_FLAGS} -fno-sign-zero -ffree-line-length-none -fallow-invalid-boz")
```

to

```
set(CMAKE_Fortran_FLAGS "${CMAKE_Fortran_FLAGS} -fno-sign-zero -ffree-line-length-none")
```

save the file and re-run the cmake and the make commands shown previously.

Once the compilation is finished, we will proceed to install lamaGOET:

```
cd ..  
git clone --recurse-submodules https://github.com/lomalaspina/lamaGOET  
cd lamaGOET/installation_July19  
git checkout periodic
```

```
sudo ./install.sh
```

In addition, we ask you to install the visualization software VESTA (also available for all 3 operating systems) which we will use for visualization of results:

<https://jp-minerals.org/vesta/en/download.html>

These should be enough to perform all the tasks in the workshop. For extra features, users can also download:

CrystalExplorer: <https://crystalexplorer.net/download/>

Orca: <https://orcaforum.kofo.mpg.de/ucp.php?mode=register>

Gaussian (paid): <https://gaussian.com/products/>

MacOS

Open the MacOS terminal and follow the instructions for Linux exchanging *sudo apt-get* by *brew*.

Examples

Remember you need to have Tonto and all other necessary softwares already installed on your machine (i.e. Gaussian), they are not provided by us, the lamaGOET is simply the graphical interface that allows the connection of those different softwares on cyclic iterations to perform the Hirshfeld Atom Refinement (HAR) among other things³.

For this workshop we have selected a few examples to run which are provided in the examples folder labeled as shown below:

Epoxide
NH3/Urea
GlyAla/L-ala
XCW
<Your_own>

If you are doing this on a windows machine: Make sure that the Xserver is running on your machine (X icon near the windows clock). If its not running, please start it by running Xming.

After starting the Xserver, start an Ubuntu session.

Run lamaGOET by simply typing lamaGOET from the terminal:

```
lamaGOET
```

This will open the graphical interface which prepare all the necessary files and run the jobs in the current directory.

The graphical interface is shown below for detailing.

³ The lamaGOET interface is still highly experimental and under development therefore bug reports are always welcome and will be corrected ASAP.

lamaGOET: An interface for quantum crystallography

Welcome to the interface for quantum crystallography

HAR | Advanced Settings for HAR | XCW | Elmodb advanced specific | Plots

Software for SCF calculation ☒ Gaussian ☐ Orca ☐ Tonto ☐ elmodb ☐ SC CC opt with Gaussian and Tonto

☐ Powder HAR ☐ Calculate .tsc file with NoSpherA2 ☐ Use Gamess for calculation of overlap integrals

Tonto executable

Jana2006 executable

Gaussian, Orca or elmodb executable

gamess_int executable

ELMO libraries folder

basis sets directory

job name(one word)

cif or pdb file ☐ Complete molecule(s) in the cif

☐ Load initial ADPs and precise coordinates from cif

hkl file ☐ write header ☐ on F ☐ on F²

Wavelength (in Angstrom) F/sigma cutoff

Charge Multiplicity

Method: Basis set

Enter manually for Gaussian, Orca or elmodb!

☐ Input external basis set manually

☐ Use Grimme dispersion (gd3bj) ☐ Use relativistic method

Extra Gaussian keywords

☐ Use SC cluster charges? SC Cluster charges radius ☐ Complete molecules ☐ Use dipoles

Refinement options (all atom types): ☒ positions and ADPs ☐ positions only ☐ ADPs only

☐ Start refinement with a Tonto IAM

☐ Refine nothing for atoms: atom labels:

☐ Refine these atoms isotropically: atom labels:

☒ Refine H positions ? ☒ Refine_H_ADPs Refine H atom isotropically?

☐ Refine anharmonic ADPs Atom labels ☐ 3rd Order ☐ 4rd Order

☐ Elongate X-H bond lengths ? if yes, enter new bond lengths below (leave empty for unchanged)

B-H: C-H:

N-H: O-H:

Apply dispersion corrections?

Number of processors available for the Gaussian or Orca job Number of processors available for Tonto

Memory available for the Gaussian, Orca or elmodb job

Figure 1 shows a preview of the lamaGOET graphic.

Example 1: Epoxide

The first example we will run is for epoxide. Navigate into the epoxide folder in your Ubuntu terminal:

```
cd /mnt/<c or d>/<Full path to folder>/epoxide
```

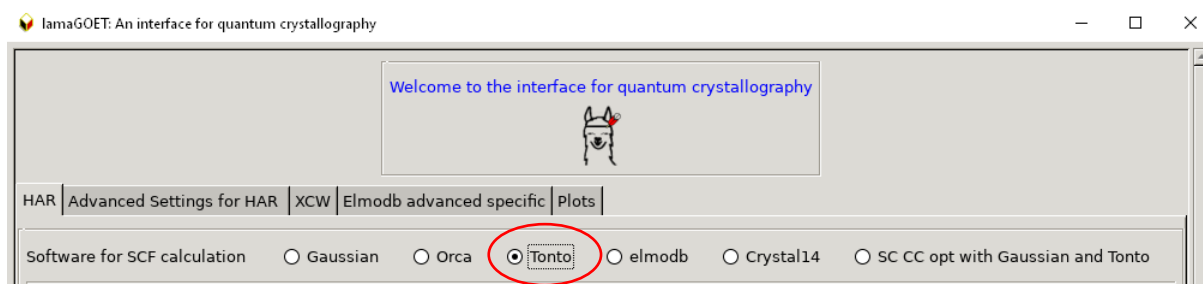
Once inside the folder, run lamaGOET by simply typing lamaGOET from the terminal inside the folder where you want to run the job (this is also the folder where all the files will be written).

```
lamaGOET
```

This will open the graphical interface which prepare all the necessary files and run the jobs in the current directory.

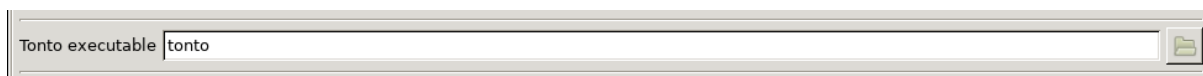
Now we need to select which QC software will be used for the calculation of the wavefunction, as we said in the beginning, lamaGOET offers you the option to use Gaussian, Orca, ELMOdb and Tonto for that step. Since not everyone will have all these softwares available we will proceed using Tonto only.

Therefore, select the Tonto in the dial selection tool like the picture below:



Once you select Tonto, some options will be automatically grayed out (or activated), so you don't need to worry about filling up all the information on the grayed-out boxes.

The next step is to tell lamaGOET the name of the Tonto executable. If you have used the install_all_you_need.sh script you don't need to change anything here. If you have not used the installation script, you just need to select the tonto executable from the "build" installation folder in your machine.



Tonto has its own set of basis sets inside its installation folder, here you just need to give the software the path to this folder, and once again, if you have used the install_all_you_need.sh script you don't need to change anything here. If you have not used the installation script, you just need to select the tonto executable from the "basis-sets" tonto folder in your machine.

basis sets directory

Now it is time to set a title for the job and give the input files:

job name(one word)

cif or pdb file ☐ Complete molecule(s) in the cif

☐ Load initial ADPs and precise coordinates from cif

hkl file ☐ write header ☐ on F ☐ on F²

You can give a job name as you wish, and then click on the folder button to select the cif file and the hkl file.

You do not need to worry about the “complete molecule(s) in the cif” button for the epoxide example, but you will see how useful it is on the next examples. The same goes for the “write header” button.

Now, please fill in the wavelength of the radiation used for this sample, and the desirable cut of criteria for F/σ (Not I/σ !) in this case it is $\text{MoK}\alpha$, therefore no changes are needed for the wavelength:

Wavelength (in Angstrom) F/σ cutoff

Extra exercise: you can run one refinement using a cutoff criteria of 3 and then another one using a cutoff criteria of zero (0.00001 in Tonto). What are the differences you see between the two results? Look at your residual density plots and fit statistics.

Now input the charge and multiplicity of your input geometry (hopefully neutral is your input is the asymmetric unit like in this case, and a closed shell system – multiplicity 1):

Charge Multiplicity

Now lets select a level of theory for the wavefunction calculation, if you are using Tonto, you can select the methods available for the dropdown menu:

Method: Basis set

Enter manually for Gaussian

☐ Input external basis set

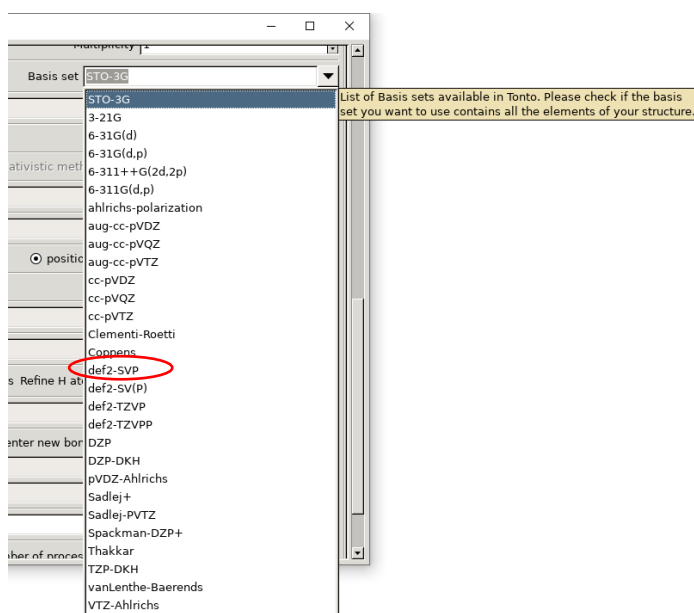
☐ Use Grimme dispersion

Extra Gaussian keywords

'rhf' - Restricted Hartree-Fock,
 'rks' - Restricted Kohn-Sham,
 'rohf' - Restricted open shell Hartree-Fock,
 'uhf' - Unrestricted Hartree-Fock,
 'uks' - Unrestricted Kohn-Sham

Notice that lamaGOET has explanation boxes that will popout if you hover the mouse on top of a button like shown above.

Do the same thing for the selection of a basis set and select **def2-SVP** from the dropdown menu:

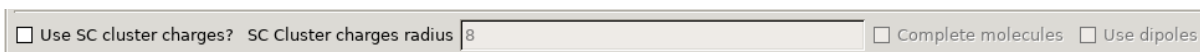


Extra: lamaGOET also offers you the option to input your own basis set manually, that is usefull for example if you want to mix different basis set for your structure, like using a heavier basis set for heavier elements and a lower basis sets for lighter elements. For that you just need to select the checkbox:

☐ Input external basis set manually

And then copy and paste basis sets from the “Basis set exchange website” into a popup window that will appear. For the sake of time, we will not demonstrate this here. If this is something you are interested in, we can do it at the end of the workshop, please let us know!

Now, we can tell Tonto to generate a cluster of point charges to simulate the crystal environment (If you don’t know what this is, please check Figure 1 in the theoretical introduction section., the “complete molecules” button here are will lead to the two different examples shown in Figure 1).



For this first example, we will not use it, so leave the box unchecked. Leaving the box unchecked means that the wavefunction calculation will be performed considering an isolated molecule (gas phase). In the next example we will see how this could affect the refinement/results.

Since Tonto uses different weighting scheme, refined against F, and allows different F/σ cutoff criteria, to be used, sometimes it is useful to start the refinement with an IAM in Tonto, this will give you the fairest comparison of the improvement of models between HAR and IAM.

☐ Start refinement with a Tonto IAM

As explained previously, Tonto does not contain any restrains, therefore we offer the following options that could be used as needed:

The screenshot shows the Tonto refinement options dialog box. It contains several sections with checkboxes and input fields:

- ☐ Refine nothing for atoms: atom labels:
- ☐ Refine these atoms isotropically: atom labels:
- ☒ Refine H positions ? ☒ Refine_H_ADPs Refine H atom isotropically?
- ☐ Refine anharmonic ADPs Atom labels: ☐ 3rd Order ☐ 4rd Order
- ☐ Elongate X-H bond lengths ? if yes, enter new bond lengths below (leave empty for unchanged)

B-H: 1.190	C-H: 1.083
N-H: 1.009	O-H: 0.983
- Apply dispersion corrections?

Which are:

- the option to keep certain atoms frozen (unrefined) during the refinement;
- the options to refine certain atoms isotropically only;
- the option to not refine any H atom positions;
- the option to elongate X-H atom bond lengths to previously known distances e.g. neutron values (this option only make sense to be used if H atoms are not refined);
- option to use dispersion correction (highly untested and not advised at the moment!!!)

For these examples we will not change any of these options, therefore, all atoms are refined anisotropically (including H atoms).

And finally, we come to the option to allocate how many CPUs and memory is available for the job (according to your machine):

The screenshot shows the Tonto resource allocation dialog box. It contains two rows of input fields and a bottom section with buttons:

- Number of processors available for the Gaussian or Orca job:
- Number of processors available for Tonto:
- Memory available for the Gaussian, Orca or elmodb job:
- Buttons: OK, Cancel

After that just click “ok” and a pop-up window will appear with the information of the option that you have input for the job and with the update of the refinement steps. You can close this window if you want (not the Ubuntu terminal).

Once the job is finished, you will see the message on the terminal that the job is finished and see how long it took (about 30 seconds?).

Search for “Begin” at the output file stdout. If you have selected to start with a Tonto IAM this block will appear twice in your output. Check the refinement statistics, see if you can locate the information listed in the table below and fill in the columns. What are the improvements from the IAM to this HAR?

At the end of every refinement, Tonto will automatically calculate a cube format file containing the residual density grid in a unit cell “cube” (therefore, not necessarily orthogonal). This can be easily visualized in VESTA.

epoxide	Shelx IAM	Tonto IAM	HAR	HAR SCCC	HAR explicit	NSA2
R(F)	0.0353					
wR(F ²)	0.0964					
$\rho_{\max}$	0.205					
$\rho_{\min}$	-0.213					
# of refl.	1308					
# of param.	44					
C-H dist.	0.997(10) 0.993(10) 0.945(11) 0.947(11)					

Now for the second example, let's try to repeat the previous refinement but this time we will simulate the crystal environment by embedding the main molecule on a self-consistent field of point charges up to a radius of 8 Å. Make a new folder called "2-epoxide", navigate into the newly created directory, and run lamaGOET.

Extra information: To avoid re-entering all the information on the GUI again, you can copy the "job_options.file" from the previous directory into the new one. This file contains a list of all the variables that you have entered into the GUI on the last job. If that file is inside the directory that you are calling lamaGOET from, the text values will be automatically re-loaded into the GUI. Be aware, that selections made on checkboxes and dropdown menus will not be loaded!

Now run lamaGOET inside this new folder, check the values in the screen and include the cluster charges by selecting the following option in the lamaGOET GUI:

Use SC cluster charges? SC Cluster charges radius 8 Complete molecules Use dipoles

After that just click "ok" and a pop-up window will appear with the information of the option that you have input for the job and with the update of the refinement steps. You can close this window if you want (not the Ubuntu terminal).

Once the job is finished, you will see the message on the terminal that the job is finished and see how long it took (about 50 seconds?).

Check the refinement statistics, see if you can locate the information listed in the table above and fill in the column (HAR SCCC). What are the improvements from the previous HAR to this new calculation? Are they significant? Why?

One last example of epoxide, we demonstrate an extra option in lamaGOET that allows you to, instead of simulating the environment with point charges, to use the actual molecules up to a certain radius! This could help improving the description of strong intermolecular interactions.

This option is activated by selecting the following option in the lamaGOET GUI:

Use explicit cluster of molecules? within radius 3 Complete molecules

Please notice that by doing that, the molecular wavefunction calculation step will be extremely slower since all the molecules are taking into account to obtain this “perturbed” description of the middle molecule. Because of that, we have already ran this job previously and give you the result files for comparison.

Notice that the time for this calculation was of about 3 hours!

Check the refinement statistics, see if you can locate the information listed in the table above and fill in the column (HAR explicit). What are the improvements from the previous two HAR models to this new one? Are they significant? Why?

For the curious ones:

Try to re-run the previous job changing the cutoff criteria of reflections from 4 (equivalent to the shelXL criteria) to 3. This will include weaker reflections in your refinement.

Check the refinement statistics, see if you can locate the information listed in the table below and fill in the columns. What are the effects of including weaker reflections? Look at your residuals.

	Shelx IAM	Tonto IAM	HAR
R(F)	0.0353		
wR(F ²)	0.0964		
$\rho_{\max}$	0.205		
$\rho_{\min}$	-0.213		
# of refl.	1308		
# of param.	44		
C-H dist.	0.997(10) 0.993(10) 0.945(11) 0.947(11)		

Now you can repeat the same step as the example before (using the cutoff criteria of 4), but now changing the level of theory, for example by changing the basis set to STO-3G. Check the refinement statistics, see if you can locate the information listed in the table below and fill in the columns. What are the effects of using this lower level of theory?

	Shelx IAM	Tonto IAM	HAR
R(F)	0.0353	0.0355	
wR(F ²)	0.0964	0.0725	
$\rho_{\max}$	0.205	0.220	
$\rho_{\min}$	-0.213	-0.225	
# of refl.	1308	1308	
# of param.	44	44	
C-H dist.	0.997(10) 0.993(10) 0.945(11) 0.947(11)	1.003(9) 0.974(8) 0.971(10) 0.958(9)	

Example 2: NH3 or Urea (user's choice)

For this example, the user can choose between urea or NH₃. The exercise here is to show how to generate the correct input when the asymmetric unit does not contain one full molecule. The reason why this requires special attention is exactly because HAR relies on a quantum chemical calculation, and therefore, a molecular fragment (ideally neutral and containing all “bonds”) must be defined/provided. The completion of “broken” fragments can be done automatically for “molecular crystals, where every atom which is closer than the covalent radius will be included in the picture. Notice please that this cannot be done in case of network compounds, as that would cause an infinite, non-stoppable growth of your system (until your computer crashes due to memory...).

For NH₃, only a third of the molecule is present in the asymmetric unit, and for urea a quarter of the molecule is present. To setup the calculation for either of the two examples, navigate into the folder, run lamaGOET, and after entering the basic info (as shown in the epoxide example 1), select the checkbox “complete the molecules in the cif” which is in from of the cif entry as shown below:



The screenshot shows a software window with a file path input field containing "/examples/inputs/nh3_third.cif" and a checkbox labeled "Complete molecule(s) in the cif" which is checked.

Notice that if you are running a Tonto IAM only it's not necessary to complete the molecule.

When filling in the information for this job, please be aware that for urea, the correct wavelength is 0.3173 Å.

run 2 different jobs for this example (each in a separate folder), one using HAR without any embedding of environment, one with point charges up to 8Å as for the epoxide example. The example with explicit cluster is provided.

Urea	Shelx IAM	Tonto IAM	HAR	HAR SCCC	HAR explicit	NSA2
R(F)	0.0253					
wR(F ²)	0.0680					
ρ _{max}	0.352					
ρ _{min}	-0.214					
# of refl.	817					
# param.	21					
N-H dist.	0.964(17) 0.900(12)					

NH3	Shelx IAM	Tonto IAM	HAR	HAR SCCC	HAR explicit	NSA2
R(F)	0.0071					
wR(F ²)	0.0191					
ρ _{max}	0.014					
ρ _{min}	-0.013					
# of refl.	98					
# param.	8					
N-H dist.	0.842(7)					

Looking at both results what can you conclude regarding the differences between IAM and HAR? Why does an explicit cluster give such different results compared to the HAR and HAR SCCC for the structure of Urea but not so much for the epoxide example? Based on that, in which cases would it make sense to use such a heavy calculation?

Example 3: GlyAla/L-Ala (user's choice)

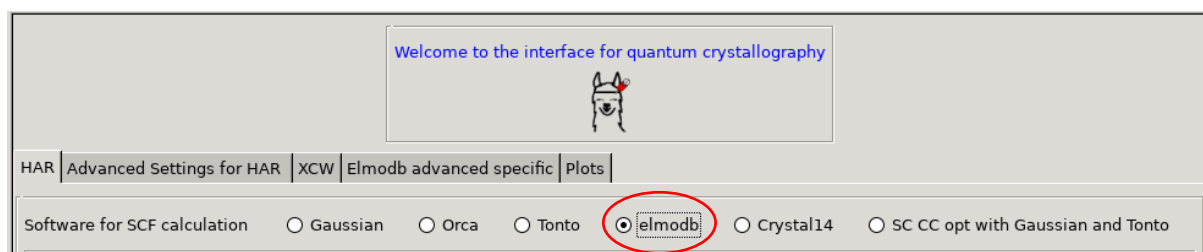
For this next example, we will demonstrate the use of the Elmo database to perform refinements using a reconstructed wavefunction based on pre-calculated Extremely Localized Molecular Orbitals database (ELMOdb). The ELMOdb program and libraries can be obtained by contacting Alessandro Genoni. For these examples we only provide the specific amino-acids and basis-set that will be used.

The libraries contain pre-calculated ELMOs for all amino-acids, making the wavefunction calculation step much faster since it only relies on a transfer from the pre-calculated database, allowing the possibility to perform HAR for macromolecules, which would be impossible otherwise.

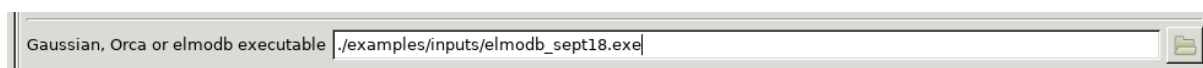
For the sake of time we will demonstrate this using the dipeptide L-glycylalanine and the aminoacid L-alanine.

Input files are provided in the 3-GlyAla and 3L-Ala folder. For the HAR-ELMO job you will need to provide a .pdb format file which contains only the atoms coordinates instead of the cif.

Let's start by selecting the ELMOdb in the list of software for the wavefunction calculation:



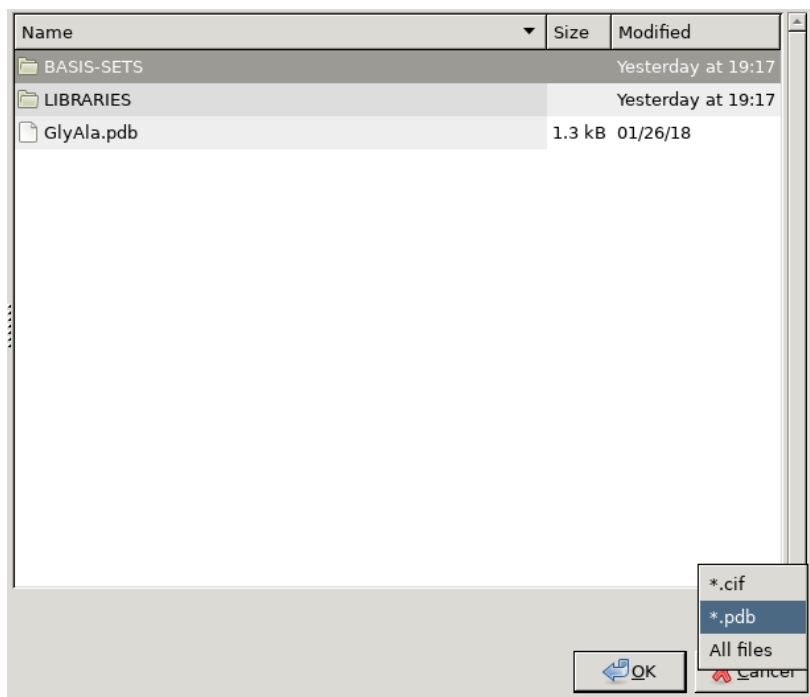
Tonto is still the software behind the partitioning and refinement, therefore, select your Tonto executable as done on the previous examples. Then using the browsing tool, locate the elmodb executable:



After that, please locate the folders for the ELMO libraries and ELMO basis -sets as shown below:



Under the input file browser, change the filter from *.cif to *.pdb so that you can see the file for selection:

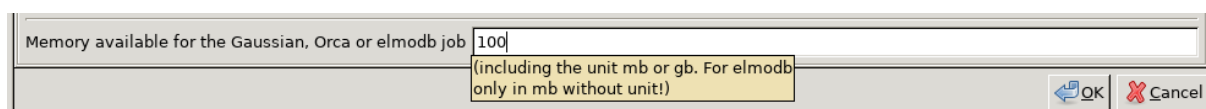


For the selection of the basis-set, you will need to enter the name of the basis-set by hand (not on the dropdown menu):



Use cc-pVDZ for GlyAla and 6-311G(d,p) for L-Ala.

For the ELMOdb, you will need to change the default definition (from 1Gb to 100 without units!) of the memory on the last menu of the GUI, this is also shown as a text info if you simply place the mouse of top of the menu:



Please notice that, when using the ELMOdb as source of wavefunction the options to use cluster charges or explicit cluster are grayed out. Could you explain why?

Now run the job and you will see on the terminal screen the information regarding all the separate steps which are running in the background:

```

Running elmodb
FORTRAN STOP
elmodb job finish correctly.
Running Tonto, cycle number 1
Tonto cycle number 1 ended
Running elmodb
FORTRAN STOP
elmodb job finish correctly.
Running Tonto, cycle number 2
Tonto cycle number 2 ended
Running elmodb
FORTRAN STOP
elmodb job finish correctly.
Running Tonto, cycle number 3
Tonto cycle number 3 ended
Running elmodb
FORTRAN STOP
elmodb job finish correctly.
Running Tonto, cycle number 4
Tonto cycle number 4 ended
Running elmodb
FORTRAN STOP
elmodb job finish correctly.
Running Tonto, cycle number 5
Tonto cycle number 5 ended
Running elmodb
FORTRAN STOP
elmodb job finish correctly.
Calculating residual density at final geometry
Job ended, elapsed time:
0 days, 0 hours, 2minutes and 44 seconds elapsed.

```

If you look at the folder where you ran the job now, you will see a structure of many separate folders. Since now you are using different software for each step, lamaGOET will always make a backup of every step of the HAR and place them inside the folder with the progression number and software. This is also the same thing that will happen if you use Gaussian or Orca for the wavefunction calculation. The collected result is concatenated in the *<jobname>.lst* file which lamaGOET creates in the same format as the stdout.

P.S.: For all effects, the *<jobname>.lst* is always the file that contains all the results when using lamaGOET, even if you select Tonto as the source of wavefunction, the result will still be written in the *<jobname>.lst* file.

Check the refinement statistics, see if you can locate the information listed in the table above and fill in the column (HAR explicit). Now try to compare the different results one could get using the different software. To what do you associate differences between Tonto_HAR and HAR-ELMO results for GLYALA and L-ALA? Why does it look like HAR-ELMO performs better for one structure and worse for the other?

GlyAla	Shelx IAM	Tonto IAM	HAR	HAR- ELMO	NSA2
R(F)	0.0243				
wR(F ²)	0.0648				
$\rho_{\max}$	0.305				
$\rho_{\min}$	-0.206				
# of refl.	2435				
# of param.	131				
N-H dist.	0.889(18) 0.899(17) 0.891(19) 0.874(17)				
C-H dist.	0.954(17) 0.946(15) 0.955(15) 0.97(2) 1.00(2) 0.973(19)				
Elapsed time	12 sec				

L-ala	Shelx IAM	Tonto IAM	HAR	HAR- ELMO	NSA2
R(F)	0.0237				
wR(F ²)	0.0637				
$\rho_{\max}$	0.348				
$\rho_{\min}$	-0.204				
# of refl.	4555				
# of param.	83				
N-H dist.	0.912(12) 0.893(11) 0.916(12)				
C-H dist.	0.950(11) 0.986(11) 0.973(11) 0.999(12)				
Elapsed time	12 sec				

Extra: Optimization with SCCC

As explained before, this is a fully theoretical calculation, where the user would provide a cif and lamaGOET will handle the optimization using either Gaussian or Orca of your structure however, embedded on a cluster of point charges. These point charges are calculated with tonto using the symmetry of the spacegroup.

This function can also be used for writing up the input files for optimization even without using the SCCC.

4- XCW - LAMA – triple bond

For this next example, we will demonstrate How to set up an XCW (or XWR = HAR + XCW). Now that you posses all the knowledge to set up a HAR, we would ask you to set up a HAR using Tonto for the structure on folder LAMA. The molecule contains one full molecule of 1-methoxy-4-[4-(4-methoxyphenyl)buta-1,3-diynyl]benzene. Set up the following HAR:

HAR:

Software: Tonto

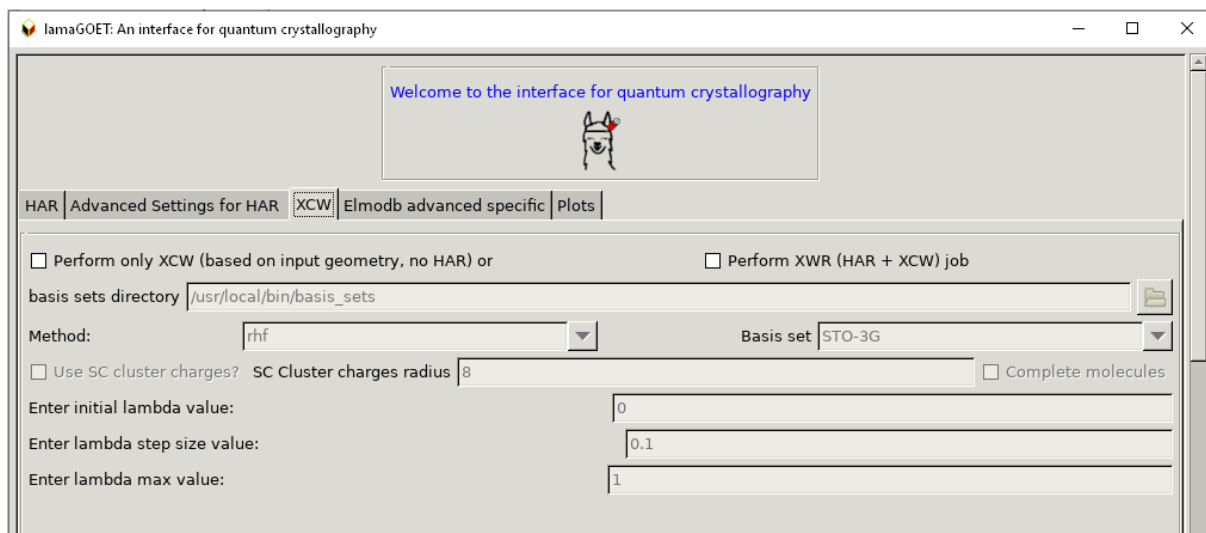
Wavelength= 1.54184

Reflection cutoff criteria= 0

Level of theory: HF/STO-3G

Using SCCC up to 8A (using complete molecules)
and starting with a Tonto IAM.

Now, before you start this job, go to the XCW tab in lamaGOET that should look like this:



Here you have the options to use different level of theory for the XCW job separately. Therefore, you could for example perform a HAR using Gaussian for the wavefunction calculation during the HAR and here you would still need to set the level of theory that will be used for the XCW in Tonto. Remember: XCW up to now, only Tonto can perform XCWs.

Select the box for performing XWR and use the following inputs:

Level of theory: HF/STO-3G

No cluster charges
Initial lambda= 0
Lambda step= 0.1
Lambda max= 0.5

Scroll down the window and click ok.

On the work folder you will see several files appearing, one for each lambda step. We will look at them using VESTA.

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3. F. L. Hirshfeld, Can X-Ray Data Distinguish Bonding Effects from Vibrational Smearing. *Acta Cryst. A* **1976**, 32, 239-244.
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5. M. Fugel, D. Jayatilaka, E. Hupf, J. Overgaard, V. R. Hathwar, P. Macchi, M. J. Turner, J. A. K. Howard, O. V. Dolomanov, H. Puschmann, B. B. Iversen, H.-B. Bürgi, S. Grabowsky, Probing the accuracy and precision of Hirshfeld atom refinement with HART interfaced with Olex2. *IUCrJ* **2018**, 5, 32-44.
6. M. Woinska, S. Grabowsky, P. M. Dominiak, K. Wozniak, D. Jayatilaka, Hydrogen atoms can be located accurately and precisely by x-ray crystallography. *Sci. Adv.* **2016**, 2, e1600192.